

OM RAJ

Software Developer • Environmental Analyst • Scientific Systems Engineer

📍 Patna, Bihar / Rupnagar, Punjab, India 📞 +91 7979734220 ✉ omraj.student@gmail.com 🌐 [omraj-website](#)

🆔 [ORCID: 0009-0002-7332-3962](#) [in linkedin.com/in/omraj](#)

📄 EXECUTIVE SUMMARY

Software developer, computational researcher, and environmental analyst with an **M.Sc. in Botany (Plant Pathology)** and project research experience at **IIT Ropar**. Specialized in building desktop software, web applications, and scientific calculation engines deployed on the **Microsoft Store** (*SNIFF DOI 2.0, WASH Suite v2.0, Calculator Luxe, Clock Span*). Expertly bridges biological research, environmental data processing (APHA 24th Ed / BIS 10500), and modern web/desktop engineering (JavaScript, HTML5, CSS3, Electron.js, Node.js, REST APIs).

</> TECHNICAL & DOMAIN SKILLS MATRIX

Software Development & Computing:

JavaScript, HTML5, CSS3, Electron.js, Node.js basics, REST API integration, Git/GitHub, Visual Studio Code.

Environmental & Water Engineering:

Wastewater testing (CETP, STP, ETP), river monitoring (pH, EC, TDS, TSS, BOD₅, COD, DO, alkalinity, coliforms, heavy metals) per APHA 24th Ed., BIS 10500, CPCB.

Agriculture & Plant Pathology:

Crop disease diagnostics, pathogen identification, IDM, stress physiology, antioxidant enzyme assays (SOD, CAT, POD, proline, MDA), biochar amendments.

Data Science, GIS & Automation:

MS Excel (advanced formulas, linear regression modeling), Power BI interactive dashboards, QGIS / basic ArcGIS, remote sensing interpretation.

Air Quality & Instrumentation:

Aerosol Spectrophotometer, TVOC Analyzer, Carbon Analyzer, Formaldehyde Analyzer, sensor calibration, air pollution spatiotemporal modeling.

Creative Design & Communication:

Adobe Photoshop, Canva Pro, Premiere Pro, CapCut, InShot, graphical abstracts for peer-reviewed journals, technical documentation, SOP authoring.

📁 PROFESSIONAL RESEARCH & SOFTWARE EXPERIENCE

Project Researcher & Scientific Systems Developer

2025 – Present

Department of Civil Engineering, IIT Ropar | Supervisors: Dr. Indramani Dhada, Dr. Neelkanth Nirmalkar | PM: Dr. Harsh Sharma

- **AWE LAB Academic Web Portal:** Architected, designed, and deployed the official research laboratory web portal for Dr. Indramani Dhada's research group at IIT Ropar.
- **Sustainable Agriculture & Biochar Project:** Evaluated comparative impact of agricultural biomass feedstocks (pineapple peel, jackfruit peel, drek wood biochars) and quantified defense enzyme kinetics (SOD, CAT, POD, Proline, MDA).
- **Wastewater & River Basin Monitoring (PPCB Project):** Conducted sampling and wet-chemical testing across CETPs, STPs, and Sutlej River basin for BOD, COD, TSS, heavy metals, and coliforms.
- **Technical Audits (Punjab Excise Dept.):** Participated in environmental audits of industrial distilleries, breweries, and bottling plants across Punjab; evaluated ZLD performance and compliance logs.

Postgraduate Dissertation Researcher — River Water Quality & Runoff

2024 – 2025

Patna University | Supervisor: Prof. Dr. Meenakshi Singh (Head, Dept. of Biotechnology)

- Executed multi-point seasonal field sampling along major ghats of Ganga River at Patna. Analyzed physicochemical and biological parameters against WHO, CPCB, and BIS 10500 standards.

Medical Writing & Scientific Research Intern

Nov 2025

YuvaIntern (in association with NSDC & Henry Harvin)

- Executed systematic literature reviews across PubMed, Scopus, Web of Science; synthesized biomedical and environmental research into technical summaries with APA, IEEE, and Vancouver citations.

Freelance Creative Designer & Digital Media Specialist

2021 – Present

Freelance & MARKAD LIMITED

- Designed graphical abstracts for journal submissions, academic branding, and video explainers. Directed media campaigns for Patna University Student Union Elections (2022 & 2025).

 EDUCATION & ACADEMIC QUALIFICATIONS

Degree / Program	Specialization / Discipline	Institution / University	Year	Score / CGPA
M.Sc. in Botany	Plant Pathology	Department of Botany, Patna University	2025	7.56 CGPA
B.Sc. in Botany	Plant Sciences & Chemistry	Patna Science College, Patna University	2023	63.50%
Diploma in Computer Applications (DCA)	Software & Computing Systems	Vinita Computer Education	2022	64.40%
Senior Secondary (Class XII, Science)	Physics, Chemistry, Biology (PCB)	Shanti Niketan Sr. Sec. School (CBSE)	2019	58.40%
Secondary School (Class X)	General Science & Mathematics	Sunshine Prep. / High School (CBSE)	2017	81.70%

 PUBLISHED SOFTWARE APPLICATIONS (MICROSOFT STORE)

- AWE LAB Academic Portal (IIT Ropar):** Official research laboratory web portal for Dr. Indramani Dhada’s research group at IIT Ropar. ([AWE LAB Portal](#))
- SNIFF DOI — Academic Literature Discovery Engine:** Published on Microsoft Store. Desktop application indexing 250M+ articles with 6–tier research gap analysis and CSL citation engine. ([Microsoft Store](#))
- WASH (Water Analysis Suite Hub):** Published on Microsoft Store. Compliance calculation engine for APHA 24th Ed & BIS 10500, spectrophotometer regression calibration, and NABL PDF export. ([Microsoft Store](#))
- Calculator Luxe & Clock Span (Watch Span):** High-precision scientific desktop tools live on Microsoft Store.

 PUBLICATIONS & RESEARCH MANUSCRIPTS

- Peer-Reviewed Journal Article:** Raj, O., & Ambreen, D. (2026). Assessment of Physicochemical Parameters of Water Samples from the Ganga River at Patna. *International Journal of Versatile Research and Analysis (IJVRA)*, 3(4), 784–810. ISSN: 2984–8903. Paper ID: IJVRA2603368.
- Research Preprint (Air Quality):** Raj, O., & Dhada, I. Spatiotemporal Assessment of Ambient Air Quality in Selected Urban, Tourist and Industrial Regions of Himachal Pradesh, India: A Case Study. *Research Square*. DOI: [10.21203/rs.3.rs-10492149/v1](#)
- Manuscript in Preparation (Sustainable Agriculture):** Raj, O., & Dhada, I. Comparative Assessment of Pineapple Peel, Jackfruit Peel, and Drek Wood Biochar on Plant Growth, Soil Fertility, and Biochemical Responses. (Evaluates SOD, CAT, POD, proline, MDA kinetics).

 WORKSHOPS & PROFESSIONAL CERTIFICATIONS

- Fundamentals of Remote Sensing:** NASA "Train – Empower – Advance" Initiative (Sept 2026).
- Advances in Genome Biology & Gene Regulation:** School of Life Sciences, JNU, New Delhi (June 2025).
- Marine Data 4 Asia 2025:** Copernicus Marine Service & Mercator Ocean International, EU Program (June 2025).
- Power BI & AI Interactive Dashboards:** OfficeMaster (Sept 2025).
- Cancer Genomics, Clinical Research & AI:** Biotechnika (Sept 2025).

 PROFESSIONAL REFERENCES

Dr. Indramani Dhada

Assistant Professor, Dept. of Civil Engineering
Indian Institute of Technology (IIT) Ropar, Rupnagar –
140001, Punjab, India
Email: indhada@iitrpr.ac.in

Dr. Raj Laxmi Singh

Assistant Professor, Dept. of Botany
Patna University, Patna – 800005, Bihar, India
Email: rajlaxmi1982singh@gmail.com